

Mangesh Bide

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PROFESSIONAL SUMMARY

Backend engineer with 2+ years of production Python and Django experience (since Feb 2024), owning the Django REST Framework and PostgreSQL backend of a HIPAA-aligned, multi-tenant healthcare SaaS on AWS (10K-100K requests/day). Led a greenfield authentication and authorization platform from architecture decision records to production (Keycloak with custom Java SPIs, OIDC, SpiceDB ReBAC), kept consistent with PostgreSQL through a transactional outbox and async worker. Shipped clinical document generation with Celery background jobs, built Terraform, Docker and GitHub Actions infrastructure, and review code for 13 engineers (**230+ pull requests**).

CORE COMPETENCIES

Python, Django & DRF

PostgreSQL & Redis

Celery Background Jobs

REST API Design

Transactional Outbox & Async Workers

AWS, Terraform & Docker

OIDC, Keycloak & ReBAC

Testing & Code Review

WORK EXPERIENCE

House Works Technology

Feb 2025 - Present

Software Development Engineer I

Remote | Promoted from intern, Apr 2025

- Own the server-side logic of a HIPAA-aligned multi-tenant healthcare SaaS on Django, Django REST Framework, PostgreSQL and AWS, handling **10K-100K requests/day**; built patient management and appointment scheduling REST APIs, an ICD-10 coding engine and AWS Cognito-based authentication
- Led a greenfield authentication and authorization platform through architecture, service integration, rollout and production deployment: self-hosted Keycloak (OIDC, custom Java SPIs, multi-tenant SSO, IdP federation) and a SpiceDB ReBAC model of **16 object types and 45 permissions** with authorization checks across the Django/DRF APIs; maintained tenant isolation across API authorization and database access
- Kept the SpiceDB authorization graph consistent with PostgreSQL using a transactional outbox and async worker, with reconciliation and consistency checks that detect and repair authorization drift
- Hardened identity: mandatory TOTP MFA for all users, instant session revocation (OIDC back-channel logout with a Redis denylist), per-tenant password policies and an audit-event producer as custom Keycloak SPIs, and user lifecycle flows (invites, suspension, termination, rehire, credential reset) recorded in an append-only audit ledger
- Shipped clinical document generation to production: a PDF template manager with clinical-note and facesheet generation, DOCX-to-PDF conversion, and bulk document export through a PDF microservice, with Celery job progress, cancellation and stale-job cleanup
- Built production infrastructure as code with Terraform (staging and production workspaces), Docker and AWS EC2, plus GitHub Actions CI/CD with automated EC2 deployments; cut deployment time from **~45 min to ~12 min (~70%)**
- Review code (**230+ pull requests for 13 engineers** since Jul 2025) and write Pytest unit and integration tests for the Django/DRF backend; migrated legacy Django views to DRF ViewSets (**~40% less boilerplate**)

Procedure Technologies

Oct 2024 - Feb 2025

Software Development Engineer Trainee

Mumbai, Maharashtra

- Selected for Procedure's "Rubber Duck" developer training cohort; built full-stack production features with React.js, Next.js, TypeScript and NestJS, including responsive interfaces with shadcn/ui
- Containerized services with Docker and provisioned cloud infrastructure with Terraform

Backend Developer Intern

Nashik, Maharashtra

- Designed REST APIs with Django and MySQL, JWT authentication and RBAC for a Flutter activity-tracking app, plus an admin panel for users, roles, tickets and activity logs
- Added ML-powered image search (Eden AI API) to an e-commerce platform, plus email-based password reset and OTP verification (Twilio)

PROJECTS

Fieldnotes: Agentic Job Research System Personal project

Agentic RAG system: a multi-stage planner-executor-synthesizer agent with tool use and embedding-based retrieval that researches companies, inspects GitHub engineering footprints, compares resume embeddings against job requirements, and returns structured apply/consider/skip verdicts with cost and latency observability.

Python, FastAPI, Gemini, Cloudflare BGE, GitHub API

HIPAA Guardian Open Source

AI-agent skill with standalone Python scanners that check source code, data, logs and API responses for all 18 HIPAA Safe Harbor identifiers, with confidence/risk scoring, audit reports, FHIR/HL7/CDA/X12 support and CI/pre-commit gates. Built with Python and Claude Code; works with Claude, Cursor, Windsurf and other AI coding agents. 6+ GitHub stars, 3 forks.

Python, Claude Code

EDUCATION

B.Tech in Computer Engineering R. C. Patel Institute of Technology

2021 - 2024

Shirpur

CGPA 7.98/10. President, Akatsuki Coding Club (Mar 2023 - Feb 2024).

Diploma in Computer Engineering R. C. Patel Polytechnic

2019 - 2021

Shirpur

93.89%

CERTIFICATIONS

AWS Certified AI Practitioner (AIF-C01)

Amazon Web Services

2026

SKILLS

Languages: Python, SQL, TypeScript, JavaScript (ES6+), Java

Backend: Django, Django REST Framework, FastAPI, NestJS, Node.js (Express), REST API Design

Databases: PostgreSQL, MySQL (relational); MongoDB, DynamoDB (NoSQL); pgvector

Distributed Systems: Transactional Outbox, Async Workers, Idempotency, Reconciliation, Event-Driven Systems

Cloud & DevOps: AWS (EC2, ECS, S3, Lambda, API Gateway, IAM, Cognito, CloudWatch), Terraform, Docker, Git, GitHub Actions, CI/CD, Shell Scripting, Oracle Cloud

Testing & Observability: Pytest, Unit/Integration Testing, Code Review, Agile/Scrum, Sentry, Grafana, ClickHouse, Structured Logging

Healthcare: HIPAA-aligned Systems, EHR Workflows, ICD-10-CM, Modifier Codes, NCCN Guidelines, FHIR and HL7 (working knowledge)

Identity & Security: Keycloak, SpiceDB (ReBAC), OAuth 2.0, OIDC, JWT, SSO, RBAC, AWS Cognito

AI / Voice: Azure Speech, RAG, Embeddings, LLM Orchestration, Prompt Engineering, Agentic AI, Gemini, OpenAI, Claude Code, Claude Skills

Frontend: React.js, Next.js, Tailwind CSS, shadcn/ui